



Project. 2

Network Infrastructure and ACL Configuration

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UNDER SUPERVISION OF

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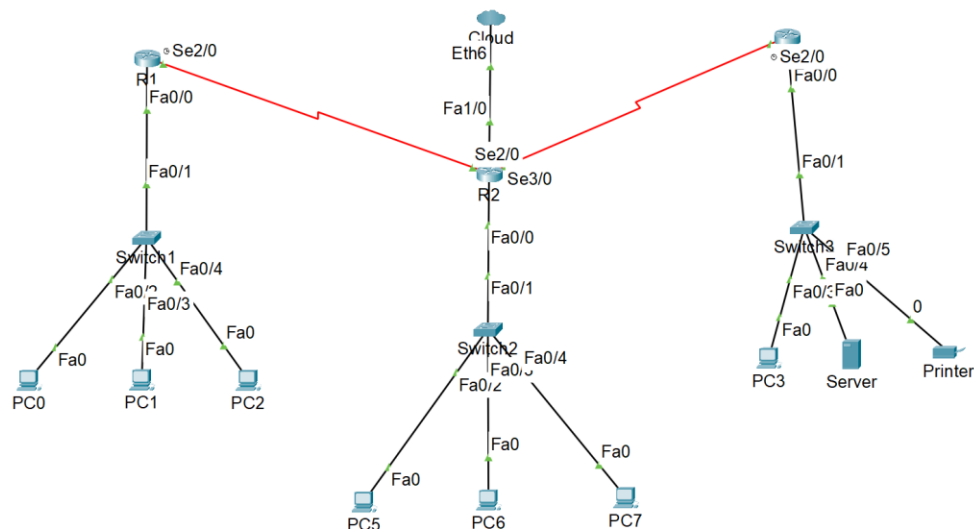
1. Introduction

This project demonstrates the design and implementation of a small network using Cisco Packet Tracer. The tasks included:

- Configuring IP addressing, routing, DHCP, and NAT.
- Implementing Access Control Lists (ACLs) to control access to the server.
- Verifying connectivity through various tests (ping, HTTP, FTP).

2. Network Configuration Summary

- **Routers:** R1, R2, R3 configured with OSPF for routing.
- **Switches:** S1 and S2 connecting PCs to the routers.
- **DHCP:** Implemented via DHCP relay for LANs.
- **NAT:** Configured on R2 to translate inside networks (10.0.0.0/24, 20.0.0.0/24, 30.0.0.0/24) to public IP (210.0.0.1).
- **Server & Printer:** Located in LAN3 (30.0.0.0/24).



3. IP Addressing

Device/Interface	IP Address	Subnet Mask	Notes
R1 F0/0	10.0.0.1	255.255.255.0	LAN (DHCP)
R1 S2/0	192.168.12.1	255.255.255.252	Link to R2

R2 F0/0	20.0.0.1	255.255.255.0	LAN(DHCP)
R2 S2/0	192.168.12.2	255.255.255.252	Link to R1
R2 S3/0	192.168.23.1	255.255.255.252	Link to R3
R2 F1/0	210.0.0.1	255.255.255.0	Link to Cloud
R3 F0/0	30.0.0.1	255.255.255.0	LAN {Static}
R3 S2/0	192.168.23.2	255.255.255.255	Link to R2
PC0	DHCP	-	Gets IP from R1
PC1	DHCP	-	Gets IP from R1
PC2	DHCP	-	Gets IP from R1
PC5	DHCP	-	Gets IP from R2
PC6	DHCP	-	Gets IP from R2
PC7	DHCP	-	Gets IP from R2
PC3	30.0.0.20	255.255.255.0	LAN {Static}
Server	30.0.0.10	255.255.255.0	LAN {Static}
Printer	30.0.0.30	255.255.255.0	LAN {Static}

4. Configurations

The following configurations were applied on the devices:

R1 Configuration

enable

configure terminal

! LAN 10.0.0.0/24

interface fastEthernet 0/0

ip address 10.0.0.1 255.255.255.0

no shutdown

! Serial link R1 ↔ R2

interface serial 2/0

ip address 192.168.12.1 255.255.255.252

clock rate 64000

no shutdown

! DHCP Pools

ip dhcp excluded-address 10.0.0.1 10.0.0.49

ip dhcp pool LAN1

network 10.0.0.0 255.255.255.0

default-router 10.0.0.1

dns-server 8.8.8.8

ip dhcp excluded-address 20.0.0.1

ip dhcp pool LAN2

network 20.0.0.0 255.255.255.0

default-router 20.0.0.1

dns-server 8.8.8.8

! Static Routes

ip route 20.0.0.0 255.255.255.0 192.168.12.2

ip route 30.0.0.0 255.255.255.0 192.168.12.2

ip route 210.0.0.0 255.255.255.0 192.168.12.2

R2 Configuration

enable

configure terminal

! LAN 20.0.0.0/24

interface fastEthernet 0/0

ip address 20.0.0.1 255.255.255.0

no shutdown

! Serial link R2 ↔ R1

interface serial 2/0

ip address 192.168.12.2 255.255.255.252

no shutdown

! Serial link R2 ↔ R3

interface serial 3/0

ip address 192.168.23.1 255.255.255.252

no shutdown

! Public Internet

interface fastEthernet 1/0

ip address 210.0.0.1 255.255.255.0

no shutdown

! DHCP Relay

interface fastEthernet 0/0

ip helper-address 192.168.12.1

! NAT/PAT Configuration

access-list 1 permit 10.0.0.0 0.0.0.255

access-list 1 permit 20.0.0.0 0.0.0.255

access-list 1 permit 30.0.0.0 0.0.0.255

ip nat inside source list 1 interface fastEthernet 1/0 overload

interface fastEthernet 0/0

ip nat inside

interface serial 2/0

ip nat inside

interface serial 3/0

ip nat inside

interface fastEthernet 1/0

ip nat outside

! ACL Rules for LAN 30.0.0.0/24

configure terminal

no ip access-list extended LAN2_FILTER

ip access-list extended LAN2_FILTER

permit tcp host 20.0.0.2 host 30.0.0.10 eq 80

permit tcp host 20.0.0.3 host 30.0.0.10 eq 21

permit icmp host 20.0.0.3 host 30.0.0.10

```
deny ip any host 30.0.0.10
```

```
permit ip any any
```

```
exit
```

```
interface serial 3/0
```

```
ip access-group LAN2_FILTER out
```

```
end
```

```
! Static Routes
```

```
ip route 10.0.0.0 255.255.255.0 192.168.12.1
```

```
ip route 30.0.0.0 255.255.255.0 192.168.23.2
```

R3 Configuration

```
enable
```

```
configure terminal
```

```
! LAN 30.0.0.0/24
```

```
interface fastEthernet 0/0
```

```
ip address 30.0.0.1 255.255.255.0
```

```
no shutdown
```

```
! Serial link R3 ↔ R2
```

```
interface serial 2/0
```

```
ip address 192.168.23.2 255.255.255.252
```

```
clock rate 64000
```

```
no shutdown
```

! Static Routes

```
ip route 10.0.0.0 255.255.255.0 192.168.23.1
```

```
ip route 20.0.0.0 255.255.255.0 192.168.23.1
```

```
ip route 210.0.0.0 255.255.255.0 192.168.23.1
```

5. ACL Rules

- **LAN2 (20.0.0.0/24):**
 - PC-5 (20.0.0.2) → Only HTTP (TCP/80) access to the server (30.0.0.10).
 - PC-6 (20.0.0.3) → Only FTP (TCP/21) and ICMP (ping) access to the server.
 - All other traffic to the server must be blocked.

Placement: The ACL was applied **outbound** on R2's Serial 3/0 interface (towards R3 and the server LAN).

- **Justification:** This ensures that PCs in LAN2 can still communicate with their gateway (20.0.0.1) for services like DHCP and ping, while filtering is applied only to traffic destined for the server.

6. Connectivity Tests

1. Output of ipconfig /all on all PCs

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::201:96FF:FE10:314E
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 10.0.0.52
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                10.0.0.1
```

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::201:C7FF:FEC3:5871
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 10.0.0.51
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                10.0.0.1
```

PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2D0:BCFF:FE70:B01E
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 10.0.0.50
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                10.0.0.1
```

PC5

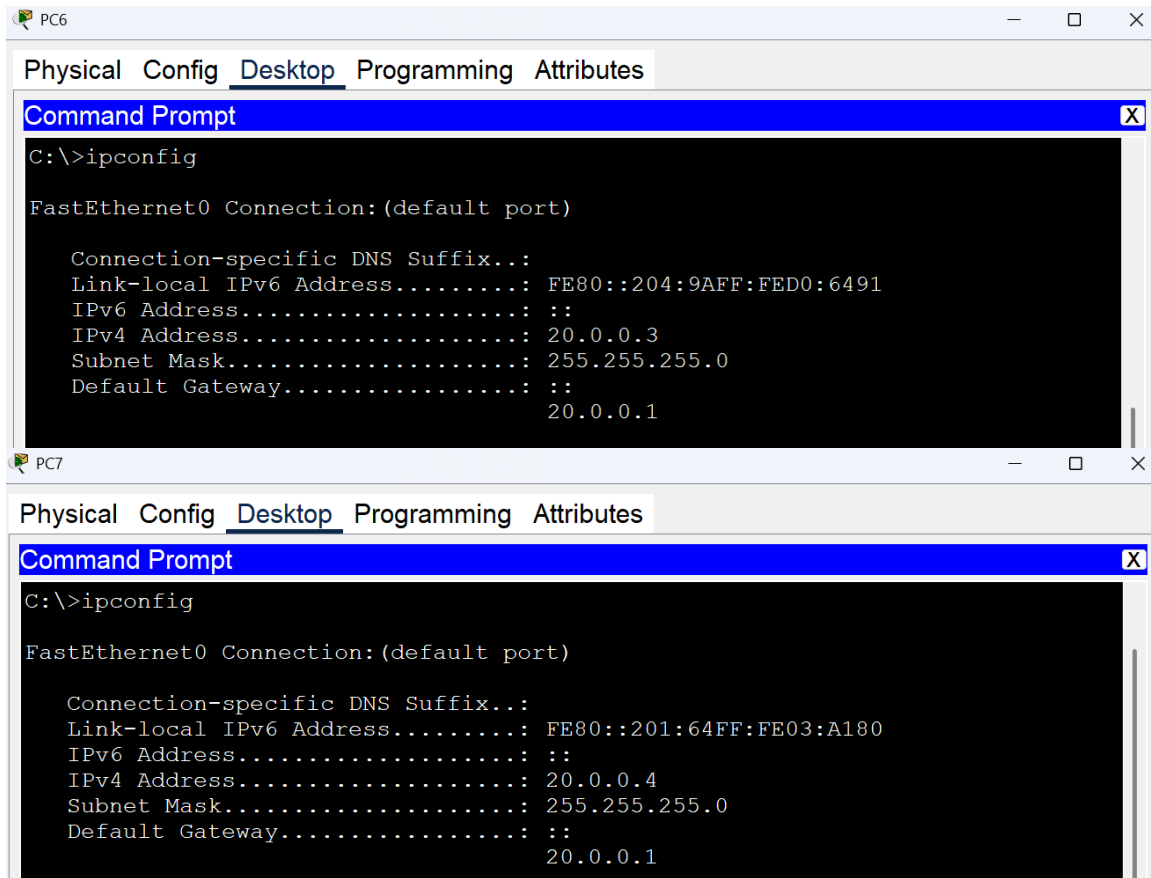
Physical Config Desktop Programming Attributes

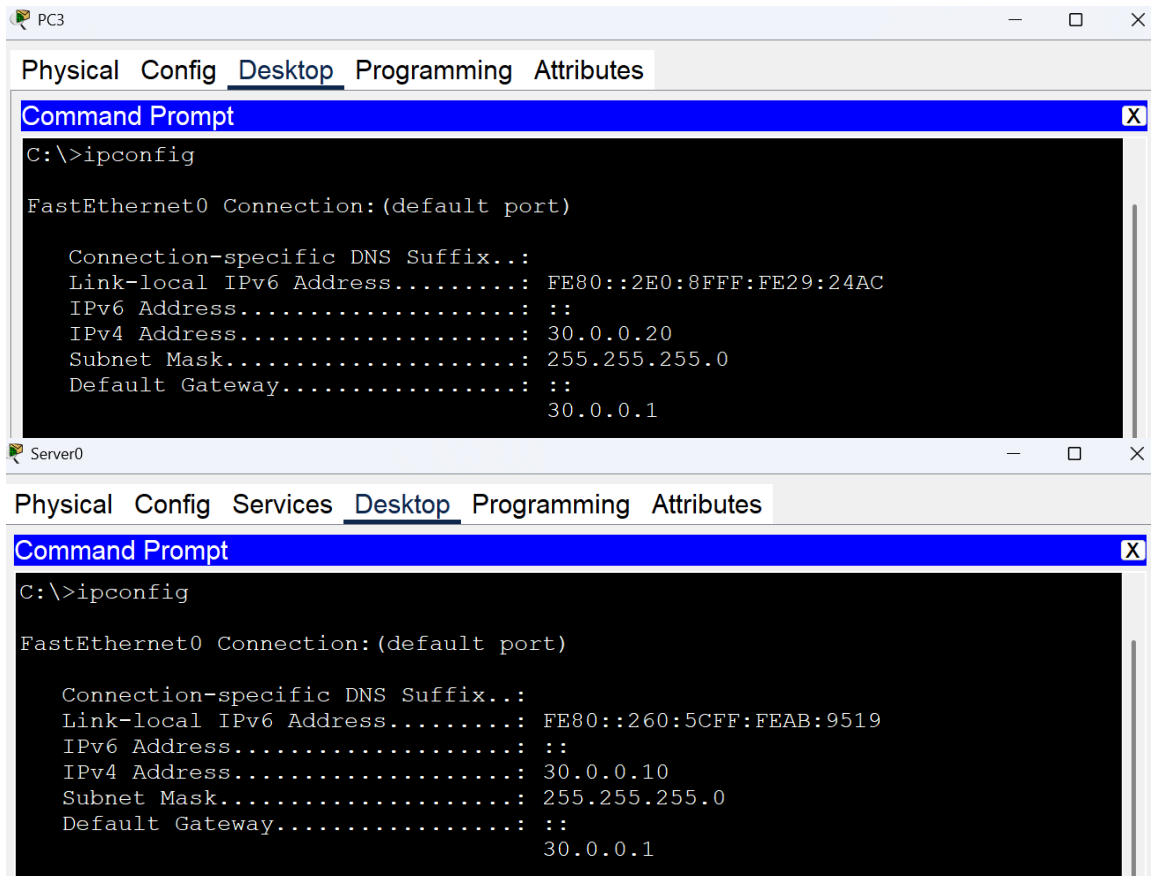
Command Prompt

```
C:\>ipconfig

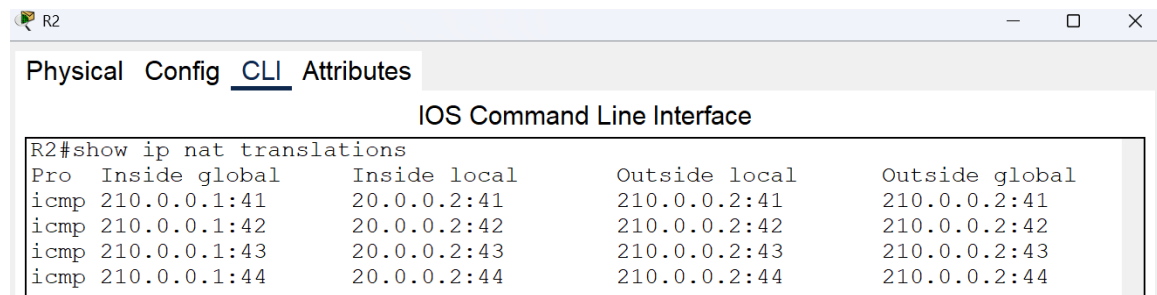
FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::204:9AFF:FEEB:1953
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 20.0.0.2
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                20.0.0.1
```



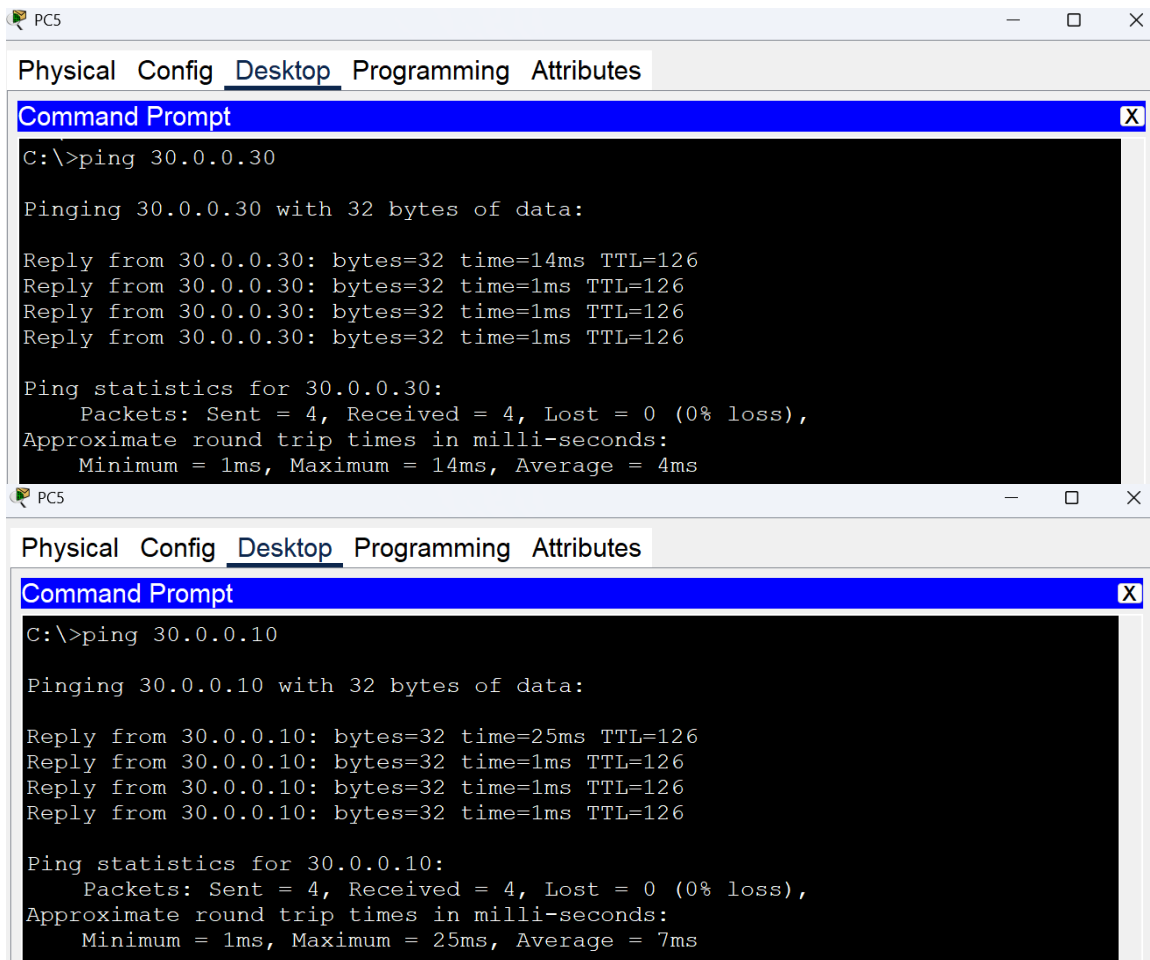


2. Output of show ip nat translations on R2

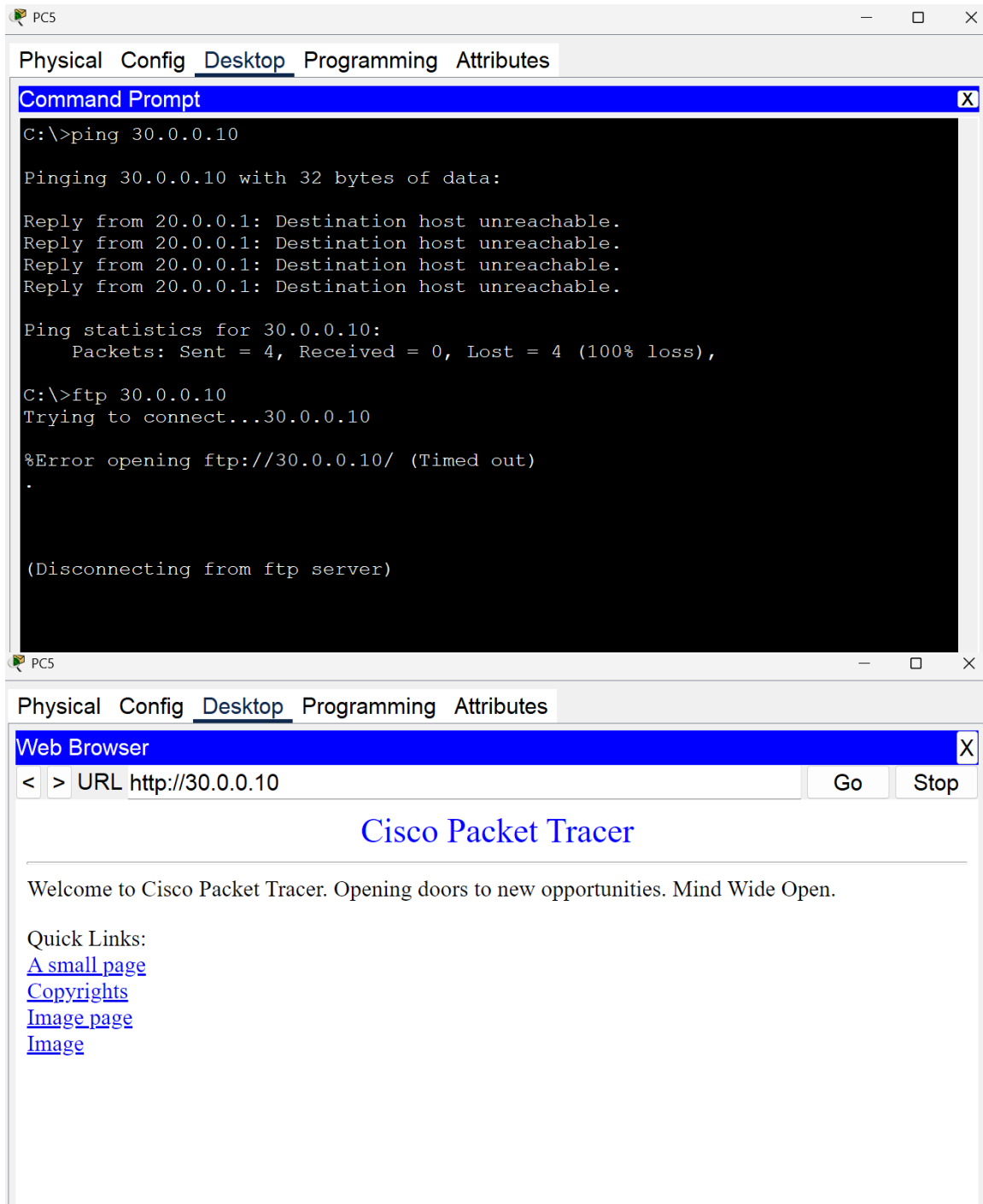


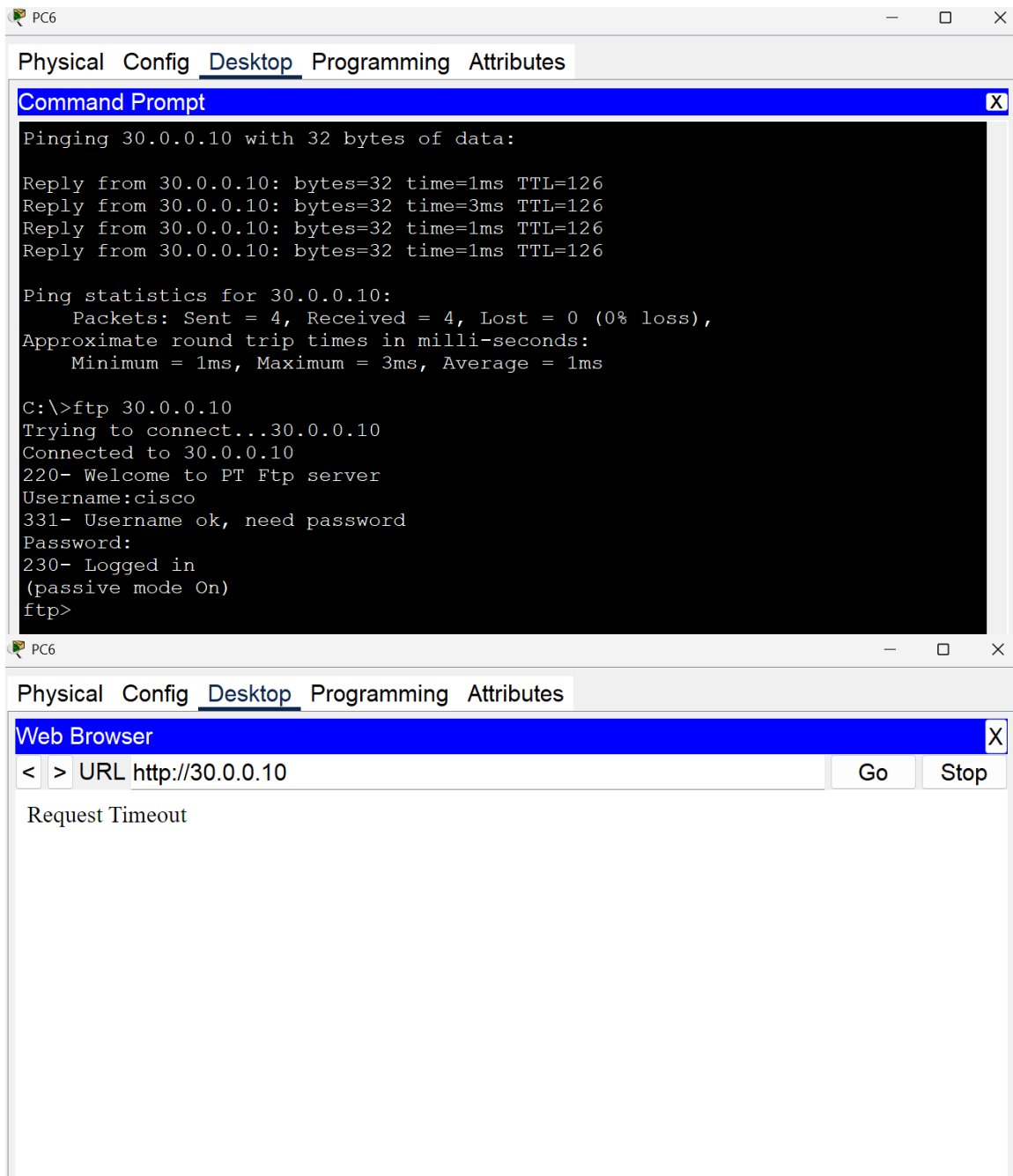
3. Connectivity tests:

- From a PC on S1: ping the printer and the server to verify reachability.



- From PCs on S2: use ping, HTTP (web browser or curl), and FTP to the server. Capture both success cases (allowed) and failure cases (blocked by ACL).





5. Conclusion

The project objectives were achieved successfully:

- The network was configured with routing, NAT, and DHCP.
- ACL rules were implemented to enforce specific access policies.
- Connectivity tests confirmed that only the allowed traffic passed through while unauthorized traffic was blocked.

This demonstrates the ability to design and secure a network using Cisco Packet Tracer.

6. Project Link

https://drive.google.com/file/d/17Tg08SS7SwXL_9OtpQ_a8GZ_jl2z3UoZ/view?usp=drive_link